Mohamed Salaheldeen

Curriculum Vitae

Institut de Mathématiques de Bourgogne (IMB), Dijon, France
mohamed-salaheldeen.mohamed@ube.fr · msalaheldn.github.io

Education

Institut de Mathématiques de Bourgogne (IMB) — Dijon, France
M2 Research Master in Mathematical Physics — Sep 2024 – May 2025
Grade: 16.8/20 (ranked 1st)
Thesis: First Contact with Factorization Homology
Advisor: Prof. Lukas Woike

The Abdus Salam International Centre for Theoretical Physics (ICTP) — Trieste, Italy
Postgraduate Diploma in High Energy Physics — Sep 2022 – Aug 2023
Thesis: 2D Topological Field Theories
Advisor: Prof. Pavel Putrov

University of Science and Technology at Zewail City — Giza, Egypt
B.Sc. in Physics, Highest Honors — Sep 2018 – Jun 2022
GPA: 3.87/4.0
Thesis: Spherically Symmetric Solutions in Modified Teleparallel Gravity
Advisor: Prof. Alexey Golovnev

Research Experience

Institut de Mathématiques de Bourgogne (IMB) — Dijon, France
Doctoral Researcher — Oct 2025 – Present
Advisor: Prof. Peter Schauenburg
Topic: Orthogonality for group-theoretical fusion categories

Bogoliubov Laboratory of Theoretical Physics (JINR) — Dubna, Russia
Research Trainee (Numerical Methods) — Sep 2021 – Nov 2021
Advisor: Prof. Yakov Shnir
Project: Numerical construction of topological solitons in scalar field theories

Center of Fundamental Physics — Giza, Egypt
Research Intern (General Relativity) — Aug 2021 – Sep 2021
Advisor: Prof. Shaaban Khalil
Project: Symmetries in Einstein field equations for the derivation of the Schwarzschild and Kerr metrics

Talks & Seminars

Invited and Contributed Talks

12 May 2026	Modular Categories and Construction of WRT Invariants (Notes) Topology Between the Waves VI, Fréjus
7 May 2026	Positive Geometry Working Group on Geometry & Physics, IMB

10 Apr 2026	The WRT 3d-TQFT from Modular Fusion Categories Working Group on (Non-)Semisimple TQFTs, IMB
10 Mar 2026	Introduction to Quantum Groups PhD Seminar, IMB

Seminar Organization

2026 – Present	Organizer, Working Group on Stable Homotopy Theory and Higher Algebra, IMB

Selected Schools & Conferences

Aug – Sep 2026	CRC 1624 Summer School: Higher Structures, Moduli Spaces and Integrability Universität Hamburg, Germany
Jun 2026	GCS '26 School: Global Categorical Symmetries Institut Henri Poincaré, Paris, France
Jun 2026	Masterclass: Seiberg–Witten Theory and 4-Manifolds University of Copenhagen, Denmark
May 2026	Spring School: Khovanov and Floer Homotopy Nantes Université, France
May 2026	Topology Between the Waves VI Fréjus, France
Mar 2026	UNIVERSE+ School: Exploring Positive Geometry Les Houches School of Physics, France
Dec 2025	Masterclass: Shifted Symplectic Structures Centre Henri Lebesgue, Angers, France
Dec 2024	Masterclass: Graph Complexes Centre Henri Lebesgue, Angers, France

Teaching Experience

Department of Applied Mathematics, Zewail City — Giza, Egypt

Teaching Assistant — Oct 2023 – Jun 2024

Courses: Ordinary Differential Equations, Partial Differential Equations, Complex Analysis, Discrete Mathematics, Numerical Analysis

Awards & Honors

2018 – 2022	Provost's Honor Roll, Zewail City (8 consecutive semesters, GPA > 3.7)
Dec 2020	Finalist, XI International Natural Sciences Tournament, Moscow, Russia
Sep 2016	National Finalist, International Physics Olympiad (IPhO), Nile University
2016	Micro Research Grant (MRG), Center of Special Studies and Programs, Bibliotheca Alexandrina

Skills

Languages	Arabic (native), English (C1), French (B1)
Computational	Python, C++, Mathematica, ROOT, LaTeX